# Harbor Route Planning Pilot

## 01. Project brief and evidence boundary

This is a fictional internal record from Harbor Parcel Cooperative. It describes Harbor Route Planning Pilot in the area of parcel dispatch. Every organization, event, participant, and number is invented for software testing. The record combines a project brief, decision notes, operational scenarios, and a handover appendix. It should be read as one bounded source rather than as an independent verification of the organization's performance.

Dispatchers rebuilt the same delivery routes in spreadsheets each morning, while depot supervisors could not distinguish a confirmed stop from a late customer amendment.

The main working role is the dispatch coordinator. The unit of work is a delivery run; the record follows its creation, review, correction, and eventual disposition. This vocabulary matters because a record can be created successfully while the larger business activity remains incomplete. A review screen, an exported file, a replayed test, and a production result describe different kinds of evidence. Later sections state which kind is actually available.

The source's status is: Measured, limited pilot. The intended audience includes the working team and the next maintainer who needs to understand a decision without reconstructing it from chat. The document records uncertainty alongside implementation detail. The final metrics section is the place to check actual observations; statements of purpose, acceptance criteria, and future plans elsewhere do not establish that those observations happened.

- Two depots and six dispatchers only
- Maximum 4,000 stops per daily import
- No driver tracking or customer location history
- Manual phone calls remain the emergency fallback

# 02. Existing workflow and the problem in practice

A typical working day begins when the dispatch coordinator receives an incomplete description of the next delivery run. The description may be technically readable but still require a decision about identity, timing, or units. The team therefore distinguished syntactic validity from business validity: being able to import a row does not mean that the row should be approved. That distinction prevents a clean-looking screen from hiding an unresolved operational question.

- Step 1: Receive a depot manifest.

- Step 2: Validate addresses and time windows.

- Step 3: Group stops into a delivery run.

- Step 4: Have a dispatcher approve the run.

- Step 5: Record exceptions during dispatch.

- Step 6: Export an end-of-day reconciliation.

The existing problem was not simply a missing database. A delivery run could be discussed in one place, changed in another, and checked later by someone who did not see the original request. The proposed record of work makes these transitions visible. It also gives the dispatch coordinator a place to leave a correction rationale instead of passing an unexplained number to the next person.

An illustrative exception is a late correction after review has begun. The first version remains part of the audit trail, while the corrected version requires its own review. The walkthrough is explanatory; it does not assert that a measured incident occurred. When this record describes an actual fictional observation, it uses the explicit incident or measurement sections rather than relying on this illustrative workflow.

# 03. Scope, exclusions, and acceptance conditions

The scope centers on delivery runs and their review history. The team deliberately avoided assigning the new component ownership of every upstream and downstream system. A useful result at this boundary is a traceable decision with a clear status, an identifiable input version, and enough context for a person to explain why the decision was made. Broader organizational effects require separate measurements.

- Two depots and six dispatchers only

- Maximum 4,000 stops per daily import

- No driver tracking or customer location history

- Manual phone calls remain the emergency fallback

Acceptance conditions were written as checks that an observer could repeat. A reviewer should be able to locate the original delivery run, inspect a corrected version without losing the first one, and distinguish a pending state from a completed state. A rejected operation should explain what information is missing. These are proposed or scoped conditions; their presence in the plan does not mean that every one passed in deployment.

The following omissions remain intentional or unresolved: Delivery travel-time savings were not measured. No production rollout beyond the two depots was approved. The sample does not establish seasonal demand behavior. A future team can investigate them, but should not fill the gaps with estimates presented as facts. The source does not establish the staffing, spending, or delivery schedule of future work. A later implementation should preserve the distinction between a requirement that was discussed and an observation that was actually recorded.

# 04. Architecture and record ownership

A browser application calls an API backed by PostgreSQL. A background worker validates CSV manifests and prepares candidate groups. Redis holds short-lived job leases; an object bucket retains the accepted input file. Drivers receive an exported run sheet rather than a new mobile application.

| Record | Important fields | Rule |
|---|---|---|
| Manifest | Depot identifier, import date, file checksum | One accepted copy per depot and business day |
| Stop | Customer code, address token, time window | A customer amendment creates a revision |
| Run | Vehicle code, depot, approval state | Only approved runs reach the driver export |
| Exception | Stop revision, reason, reporter | Exceptions never overwrite the original request |
| Import job | Source checksum, row counts, status | Rejected rows remain inspectable |
| Audit event | Actor, action, business day | Append-only event history |

The ownership boundary is more important than the technology label. The component that accepts a delivery run owns the validation and status history for that record; it does not automatically become the source of truth for a downstream business result. A maintainer should be able to identify both the original evidence and the decision made from it. Private working material is retained only within the intended test scenario.

A schema migration should preserve old identifiers and the meaning of earlier statuses. Replaying an earlier input against newer code can be useful for diagnosis, but the replay is a new observation with its own conditions. It must not silently rewrite the recorded history of the original attempt. The fictional record therefore keeps input identity, processing state, and human review as separate concepts.

# 05. Decision log and rejected alternatives

| Decision | Recorded rationale |
| --- | --- |
| Explicit dispatcher approval | The grouping rule is advisory; a person owns dispatch decisions. |
| Business-day import key | A checksum alone cannot identify a corrected depot manifest. |
| Append-only amendments | Reconciliation must explain which stop revision was actually dispatched. |
| No live traffic integration | The pilot evaluates the workflow without promising road-time optimization. |

The first alternative considered was allowing the dispatch coordinator to overwrite the latest record without keeping prior values. It was easy to describe, but would have made it difficult to explain which version a colleague had reviewed. The decision notes favor a visible correction path instead. The choice adds a review step, yet preserves the evidence required to diagnose disagreements later.

The second alternative was treating every successful technical operation as business completion. For example, receiving a response or producing an export says that one step finished; it does not prove that all delivery runs reached their intended business destination. The team documented this distinction so that progress updates would not grow into unsupported success claims.

A later team may choose a different implementation if its constraints change. The useful part of this log is the reasoning: The grouping rule is advisory; a person owns dispatch decisions. Reconciliation must explain which stop revision was actually dispatched. Those statements explain why the original boundary existed. A replacement should either preserve the same assurance or explicitly record why a different tradeoff is acceptable.

# 06. Working chronology and review notes

These fictional working dates organize the team's discussion of parcel dispatch. They are not deployment receipts. Several activities can occur while a project is still proposed or limited to a controlled environment. Read a date next to its evidence classification rather than assuming that chronological progress implies a launch.

| Date | Working note | Evidence classification |
|---|---|---|
| 2026-02-03 | Working session examined: receive a depot manifest. The note records the question and its required evidence; it does not certify a live release. | Design or review note |
| 2026-02-06 | Working session examined: validate addresses and time windows. The note records the question and its required evidence; it does not certify a live release. | Design or review note |
| 2026-02-09 | Working session examined: group stops into a delivery run. The note records the question and its required evidence; it does not certify a live release. | Design or review note |
| 2026-02-12 | Working session examined: have a dispatcher approve the run. The note records the question and its required evidence; it does not certify a live release. | Design or review note |
| 2026-02-15 | Working session examined: record exceptions during dispatch. The note records the question and its required evidence; it does not certify a live release. | Design or review note |
| 2026-02-18 | Working session examined: export an end-of-day reconciliation. The note records the question and its required evidence; it does not certify a live release. | Design or review note |

The review format asked three questions at each transition: what input was available, what decision was made, and what remained unknown? For delivery runs, that prevented missing source material from being concealed by a status update. A note saying that the team examined a scenario only establishes the examination; an observed test result needs an explicit environment and expected behavior.

Meeting notes were kept short enough for the dispatch coordinator to revisit. Where two notes disagree, the disagreement remains part of the source. A later summary must not select the more optimistic statement merely because it appears later in the file. Closure depends on the evidence and its scope, not on the position of a paragraph.

# 07. Detailed walkthrough of one unit of work

### *Scenario 1: Receive a depot manifest*

In this illustrative scenario, the dispatch coordinator works with the manifest record while handling a delivery run. The visible information includes depot identifier, import date, file checksum. The governing rule is: One accepted copy per depot and business day. If required evidence is absent, the person leaves the operation pending and records the missing condition. This scenario describes intended handling, not a separately measured success rate.

### *Scenario 2: Validate addresses and time windows*

In this illustrative scenario, the dispatch coordinator works with the stop record while handling a delivery run. The visible information includes customer code, address token, time window. The governing rule is: A customer amendment creates a revision. If required evidence is absent, the person leaves the operation pending and records the missing condition. This scenario describes intended handling, not a separately measured success rate.

### *Scenario 3: Group stops into a delivery run*

In this illustrative scenario, the dispatch coordinator works with the run record while handling a delivery run. The visible information includes vehicle code, depot, approval state. The governing rule is: Only approved runs reach the driver export. If required evidence is absent, the person leaves the operation pending and records the missing condition. This scenario describes intended handling, not a separately measured success rate.

### *Scenario 4: Have a dispatcher approve the run*

In this illustrative scenario, the dispatch coordinator works with the exception record while handling a delivery run. The visible information includes stop revision, reason, reporter. The governing rule is: Exceptions never overwrite the original request. If required evidence is absent, the person leaves the operation pending and records the missing condition. This scenario describes intended handling, not a separately measured success rate.

### *Scenario 5: Record exceptions during dispatch*

In this illustrative scenario, the dispatch coordinator works with the import job record while handling a delivery run. The visible information includes source checksum, row counts, status. The governing rule is: Rejected rows remain inspectable. If required evidence is absent, the person leaves the operation pending and records the missing condition. This scenario describes intended handling, not a separately measured success rate.

### *Scenario 6: Export an end-of-day reconciliation*

In this illustrative scenario, the dispatch coordinator works with the audit event record while handling a delivery run. The visible information includes actor, action, business day. The governing rule is: Append-only event history. If required evidence is absent, the person leaves the operation pending and records the missing condition. This scenario describes intended handling, not a separately measured success rate.

# 08. Incident, exception, and recovery record

On pilot day four, a corrected manifest arrived after a dispatcher had approved its first revision. A draft export briefly mixed two revisions. The team disabled the draft export, kept the already approved run sheet available, and added a revision check at approval and export. No completed delivery was asserted to have been lost.

The immediate response separates the current operational question from a permanent fix. The dispatch coordinator needs to know which delivery runs are affected, which original inputs remain available, and whether continuing would make reconciliation harder. A workaround can restore a narrow workflow while leaving the underlying design question open. That limited restoration should not be reported as proof that the whole component is reliable.

The follow-up record should name the scenario, identify its input version, and state the boundary of any retest. An entry that passes once is a useful observation but does not certify every source format, every operating condition, or every future change. Where this fictional archive has no such result, its absence remains visible rather than being filled by a generalized claim.

# 09. Observed results, targets, and uncertainty

Across ten observed business days, median route preparation time fell from 46 minutes to 29 minutes for the two pilot depots. Seventy-six of 80 sampled run sheets reconciled without a manual correction; the other four required address amendments. These are pilot observations, not company-wide results.

| Measure | Starting point or scope | Recorded value | Interpretation limit |
|---|---|---|---|
| Median preparation time | 46 minutes | 29 minutes | Ten days; two depots |
| Run sheets without correction | Not measured before pilot | 76 of 80 | Address amendments counted as corrections |
| Customer satisfaction | Not measured | Not measured | No survey was conducted |

Interpretation must preserve the environment and denominator attached to a result. A count of delivery runs in a rehearsal cannot be presented as the number completed in production. A time comparison does not establish a separate quality improvement. If an earlier section describes a goal, that goal remains a goal until this section contains a compatible observation with clear scope.

The unresolved items are: Delivery travel-time savings were not measured. No production rollout beyond the two depots was approved. The sample does not establish seasonal demand behavior. None of these gaps is resolved by the amount of technical detail in the architecture section. A reader can understand a design well while still being unable to establish its operational outcome. If a summary needs to be shorter, keep the uncertainty rather than removing it to make the project sound complete.

# 10. Documented lessons and reusable boundaries

- Bind approval to a specific manifest revision.

- Keep the manual export path available during worker incidents.

- Measure preparation time separately from delivery performance.

The reusable part is the relationship between a delivery run, its original evidence, and the review decision. Copying an implementation without that relationship could reproduce the interface while losing the reason that the team could trust a particular status. A future maintainer should ask whether the same identity rules, review authority, and correction history still apply in the new setting.

Reuse also needs a boundary statement. A checksum alone cannot identify a corrected depot manifest. That rationale is tied to this project's constraints; it is not a universal rule that replaces investigation. Before adapting the design, compare the new inputs, downstream authority, failure recovery requirements, and evidence needed to claim completion. Differences should be recorded as open decisions rather than hidden inside a copied configuration.

The lessons listed here were explicitly recorded in this fictional source. They are not claims that every recommendation was implemented successfully. In particular, a lesson drawn from a rehearsal or rollback can be valuable even if the project has no positive production outcome. Keep the lesson's origin visible when summarizing it for another team.

# 11. Handover, access, and operational questions

A person taking over parcel dispatch work should first locate the current status of each delivery run and the original material supporting that status. The handover should distinguish ordinary operation, pending review, and an exception that needs a named decision. Access to a review screen does not grant authority to change a downstream system or certify the final business result.

- Find the original manifest input before changing a derived field.

- Check the run identity and the effective version used by the review.

- Ask the dispatch coordinator to explain the normal correction path.

- Record the evidence scope when handing an issue to another person.

- Preserve a pending state when the required approval or source is absent.

- Treat environment access and data retention as separate questions from project success.

The archive intentionally does not include credentials, personal production records, or an executable deployment procedure. Environment names in this report identify fictional evidence contexts. They should not be converted into real infrastructure settings. A reader testing a document-processing application can use the text to check whether uncertainty survives summarization without trying to operate the fictional service.

Open handover questions remain concrete: who can authorize a correction, which record version is being reviewed, and what event proves that the next boundary accepted it? Delivery travel-time savings were not measured. No production rollout beyond the two depots was approved. Those missing answers matter more than a confident closing paragraph. They remain explicit so another reader can decide what evidence to collect next.

# 12. Evidence index and final source notes

| Source marker | What this report actually contains |
| --- | --- |
| 01-BRIEF | Dispatchers rebuilt the same delivery routes in spreadsheets each morning, while depot supervisors could not distinguish a confirmed stop from a late customer amendment. |
| 01-ARCH | A browser application calls an API backed by PostgreSQL. A background worker validates CSV manifests and prepares candidate groups. Redis holds short-lived job leases; an object bucket retains the accepted input file. Drivers receive an exported run sheet rather than a new mobile application. |
| 01-INCIDENT | On pilot day four, a corrected manifest arrived after a dispatcher had approved its first revision. A draft export briefly mixed two revisions. The team disabled the draft export, kept the already approved run sheet available, and added a revision check at approval and export. No completed delivery was asserted to have been lost. |
| 01-RESULT | Across ten observed business days, median route preparation time fell from 46 minutes to 29 minutes for the two pilot depots. Seventy-six of 80 sampled run sheets reconciled without a manual correction; the other four required address amendments. These are pilot observations, not company-wide results. |
| 01-GAPS | Delivery travel-time savings were not measured. No production rollout beyond the two depots was approved. The sample does not establish seasonal demand behavior. |

This index helps a reviewer locate the source of a statement; it is not an additional outcome. The brief describes the problem, the architecture describes component responsibilities, and the incident record describes an exception with its evidence boundary. The results entry is the only consolidated statement of the fictional observations. All quotations should refer to the words actually present in this version of the report.

A future amendment should keep the old record available and say which claim it corrects. For Harbor Parcel Cooperative, that means a new note cannot silently replace the scope of the delivery runs already described here. If later evidence resolves an unknown, the new evidence needs its own date and source. No such later amendment is included in this file.

End of the fictional source record. Current status: Measured, limited pilot. The report ends with the same boundary it began with: a technical description is useful evidence about the design, while an operational result must be grounded in an explicit observation. Any stronger conclusion requires additional source material.